

World's Smallest Programmable INDUSPHONE DESIGN

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Smartphones have become a necessity in daily life. Smartwatches also help in staying connected with phones and receiving notifications. Call access is available directly through smartwatches. Such designs have been published in EFY. A DIY article on world's smallest phone with e-paper was previously published in EFY (<https://www.electronicsforu.com/electronics-projects/e-ink-phone-diy-project>). A smartwatch DIY article was also published (<https://www.electronicsforu.com/electronics-projects/super-innovative-projects/smartwatch-with-indusboard>).

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In continuation of these designs, the world's smallest phone design has now been developed. This smartphone, just 3cm in length, is somewhat the size of a smartwatch. Since the design is somewhat complex and lengthy, it will be published in multiple parts, allowing step-by-step guidance for better understanding.

Designing

As this is to be the world's smallest smartphone, the components must be selected carefully to fit within its small frame. The IndusBoard coin

was chosen due to its compact size (around 3cm) and because it includes many I/Os required for this device. It also integrates inbuilt sensors, such as a magnetometer, accelerometer, temperature sensor, and Wi-Fi, all within a 3cm size, helping to keep the device compact. A display of the same size with a touchscreen is also required.

Before proceeding with the design, the smartphone's functions need to be outlined. It supports 2G and 4G SIM cards, Wi-Fi, and a touchscreen display. The basic phone functions include calls, messaging, contacts, dialling, and loudspeaker mode.

Additional features, such as games, a calculator, Wi-Fi contact



Fig. 1: IndusPhone smartphone calling screen

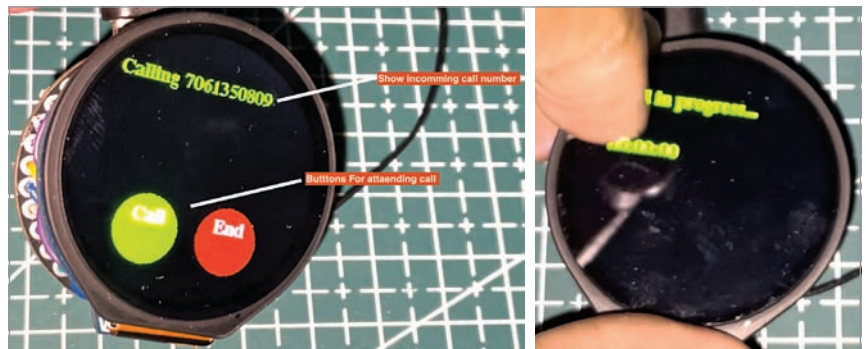


Fig. 2: Incoming call screen UI design

BILL OF MATERIALS

ID	Name	Designator	Footprint	Quantity	Manufacturer Part	Manufacturer
1	CST816 Touch Round Display Glass	CN1	CST816	1	CST816	
2	GSM Module	M1	SIM800LMODULE_SIM800LMODULE	1	800I	SIM Com
3	MIC14	M2	MICRO-USB-SMD-KH-MICRO0.8CB-5PJ	1		
4	SMD Speaker	SP1	SPEAKER	1	GSPK151035PN-8R0.5W-L35-1.25T	INGHAI
5	IndusBoard Coin	U1	INDUSBOARD_COIN	2	IndusBoard COIN	IndusBoard
6	GC9A01	U3	GC9A01	1	Waveshare	
7	SMC-204-ARP6	U4	SIM-SMD_SIM-SMD_MICROSIM_SMC-204	1	SMC-204-ARP6	XUNPU